BALAJI H

SOFTWARE ENGINEER | FULL STACK DEVELOPER — MERN | AI INTEGRATION

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PROFESSIONAL SUMMARY

Full-Stack Developer specializing in secure, scalable, and performance-optimized web applications using the MERN stack. Built and deployed three full-stack applications, including an AI-powered telemedicine platform achieving ~400ms average API response time and an 89 Lighthouse performance score, and a multi-tenant SaaS platform with real-time anomaly detection. Proficient in React.js, Node.js, Express.js, MongoDB, REST API development, JWT/RBAC authentication, Socket.IO, and Google Gemini API integration. Transitioned from Biotechnology to software engineering through hands-on learning and project development, applying a skills-first approach along with strong analytical, problem-solving, debugging, and testing abilities to build reliable, production-oriented applications.

INTERNSHIP

Software Engineer Intern (Full-Stack Development) — Codomax Digital Solutions **July 2026 – Aug 2026**

- Developed full-stack features for a Hospital Management System using React.js, Node.js, Express.js, and MongoDB, building and integrating RESTful APIs for managing patients, doctors, appointments, and hospital workflows.
- Implemented JWT authentication and Role-Based Access Control (RBAC) to secure application access and backend endpoints.
- Developed responsive React.js components and reusable UI modules to improve user experience and maintainability.
- Integrated AI-powered features into application workflows and performed API testing, debugging, database operations, and performance optimization while following Agile development practices.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL
Frontend: React.js, Redux Toolkit, Vite, HTML5, CSS3, Ant Design
Backend: Node.js, Express.js, Spring Boot, REST API Design
Database & Caching: MongoDB, MySQL, Redis
Auth & Security: JWT, Role-Based Access Control (RBAC)
Real-Time: Socket.io, WebSockets
DevOps: Git, GitHub Actions (CI/CD), Vercel
Testing: Jest, Supertest, Vitest, React Testing Library
AI Integration: Google Gemini API

PROJECTS

CareSync — AI-Powered Doctor Appointment System **June 2026 – Aug 2026**

GitHub | Live Demo

Tech Stack: React.js, Redux Toolkit, Node.js, Express.js, MongoDB, JWT, RBAC, Google Gemini API

- Developed a full-featured platform supporting patient, doctor, and admin workflows for appointment booking, records, and prescriptions.
- Designed optimized REST APIs achieving an average API response time of 400ms, verified via Chrome DevTools Network panel.
- Achieved an 89 Lighthouse Performance score with a Largest Contentful Paint of 0.82s through code splitting and lazy loading.
- Integrated Google Gemini API to deliver AI-assisted features and real-time telemedicine capabilities.
- Implemented a CI/CD pipeline via GitHub Actions with automated deployment to Vercel's serverless infrastructure, ensuring fast global load times and zero-downtime updates.
- Ensured application reliability with unit and integration tests using Jest, Supertest, Vitest, and React Testing Library.

InsightOps — AI-Powered Inventory & Expense Management Platform **March 2026 – April 2026**

GitHub | Live Demo

Tech Stack: React.js, React Query, Context API, Node.js, Express.js, MongoDB, Redis, Socket.io, JWT, RBAC, Google Gemini API

- Built a multi-tenant SaaS platform ensuring complete data isolation across organizations using scoped middleware and centralized state management via React Query and Context API.
- Implemented real-time anomaly detection using Z-score statistical analysis to flag unusual expenses for admin review.
- Integrated Google Gemini API for natural language search, converting plain English queries into secure, sandboxed database queries with dynamic table/chart results.
- Used Socket.io to power real-time notifications for anomalies and inventory changes without requiring a page refresh.

ReserveX — Premium Ticket Booking Platform **Oct 2025 – Dec 2025**

GitHub

Tech Stack: React.js, Redux, Node.js, Express.js, MongoDB, Socket.io, JWT, RBAC

- Built a MERN ticket booking platform with real-time seat locking via Socket.io and atomic concurrency control to prevent double-bookings.
- Implemented three-tier RBAC (User, Manager, Admin) with secured REST APIs, a custom promo engine, and Recharts-based analytics dashboards.
- Designed a responsive, glassmorphic UI with Google Sign-In, GPS-based location detection, and UPI QR code checkout.

EDUCATION

B.Tech – Biotechnology, Bannari Amman Institute of Technology **Graduated 2026**
CGPA: 8.0 / 10

Senior Secondary (XII), SRV Matric Hr. Sec. School **2021 – 2022**
Score: 85%

Secondary (X), SRV Matric Hr. Sec. School **2019 – 2020**
Score: 96%

CERTIFICATIONS

- Agentic AI Certified Foundations Associate — Oracle, July 2026
- Full Stack Development — GUVI x HCL, May 2026

ACHIEVEMENTS & LEADERSHIP

- Winner, SRM TechSpectrum '25 — recognized for full-stack project excellence.
- Mentored 50+ students in React and Java as a Technical Trainer.
- Served as Event Head for LEO-FEST and AURA'25, leading end-to-end technical operations including logistics, AV/technical setup, and cross-team coordination.